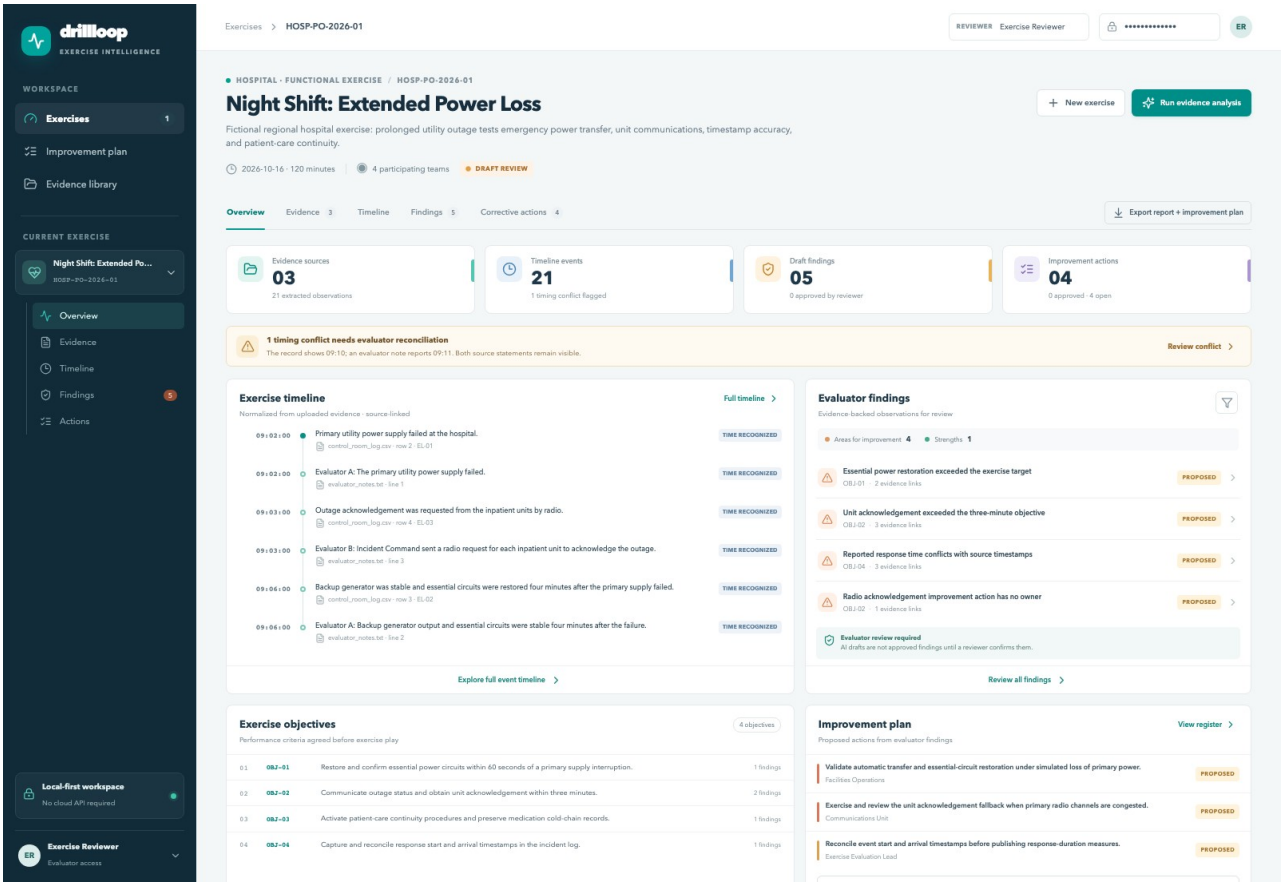
EUREKA / PRODUCT CONCEPT

DrillLoop

Turn exercise evidence into findings people can review and tasks teams can verify

A local-first review workspace for emergency preparedness teams, hospitals, airports, civil defense, training providers, and critical infrastructure operators.

Product thesis
Turn fragmented exercise evidence into a source-linked timeline, evaluator-reviewed findings, and corrective actions that remain open until an authorized reviewer checks closure evidence.



DrillLoop hospital demo · interface screenshot · all records are synthetic

Prepared 27 October 2026 · Hackathon MVP concept and implementation

Start here: one-minute example

In the fictional hospital exercise, the control-room log records a power failure at 09:02 (EL-01) and essential power restored at 09:06 (EL-02). The exercise target is 60 seconds.

What DrillLoop proposes
The recorded restoration took about four minutes, longer than the one-minute target. Evidence: EL-01 and EL-02 in control_room_log.csv. The cause is not in these records, so it stays unknown. The finding remains proposed until an evaluator reviews it.

Source says	System suggests	Person decides
Power failed at 09:02; stable at 09:06.	Possible improvement gap against OBJ-01.	Accept, edit, or reject the proposal; assign an owner only when known.

For a glossary, audience-specific reading paths, and a conceptual example for AI agents, see DRILLLOOP_READER_GUIDE.md.

1. Executive summary

Emergency exercise evaluators turn notes, event logs, recordings, and observation forms into an After-Action Report and Improvement Plan (AAR/IP). Evidence is often scattered across files, timing conflicts need review, and improvement tasks can lose ownership or completion proof. DrillLoop keeps each proposed finding connected to its source record and leaves decisions with the human evaluator.

The working MVP imports notes, CSV event logs, JSON transcripts, and PDFs with selectable text. It builds a timeline from recorded times, proposes a small set of findings for the fictional hospital exercise, supports evaluator approval or rejection, tracks owners and completion evidence, and exports an AAR/IP PDF.

The default analysis uses fixed local rules and needs no cloud service. An optional local language model can reword an action draft but cannot create findings or citations. An optional speech-to-text path needs a model installed on the computer. Neither optional model was used in the reported benchmark.

Problem and target user

The main user is an exercise lead or evaluator who must reconcile observations against pre-stated objectives and prepare a reviewer-ready report. Action owners need measurable tasks; senior preparedness reviewers need a defensible audit trail. Target customers include municipal emergency teams, hospitals, airports, civil defense, training providers, defense-support training organizations, and critical infrastructure operators.

Why now / product value to test

The value hypothesis is that preserving source-level traceability and keeping corrective actions tied to evidence and reviewer decisions can reduce evidence hunting and make follow-up more reliable. No human time-savings study or ROI claim has been measured. The pilot must show whether review workload, rework, and closure verification improve in a real evaluator's workflow.

2. Workflow and practical demo scenario

Step	Evaluator workflow	DrillLoop output
Plan	Set scenario, date, targets, criteria, roles, teams, and milestones.	Planning record connects findings to the targets being tested.
Collect	Upload notes, event logs, transcripts, and reports.	Original file, file-change checksum, extracted observations, and source locations.
Reconstruct	Put recorded times in order and compare sources.	Missing or conflicting times stay marked as uncertain.
Review	Read the draft and its supporting records; edit, approve, or reject.	The reviewer's decision is recorded; proposals remain distinct from approvals.
Improve	Assign an owner, due date, and a test for completion.	The action cannot be approved until its finding is approved and required fields exist.
Verify	Check the completion proof before closure.	A reviewer closes the task only after checking evidence.

Fictional hospital prolonged power outage

Exercise HOSP-PO-2026-01 ("Night Shift: Extended Power Loss") tests four objectives. Included evidence is synthetic and deliberately covers failure, improvement, a positive performance example, conflicts, and missing details.

Time / signal	Synthetic source statement	Expected treatment
09:02	Primary utility power supply failed.	Source-backed event; link to OBJ-01.
09:06	Backup generator stabilized and essential circuits restored four minutes later.	Proposed gap against the 60-second objective.
09:03 → 09:10	Radio acknowledgement request; ICU acknowledgement arrived by runner after seven minutes.	Proposed communications finding; evaluator also reports 09:11, unresolved.
09:02 → 09:08	Outage start and facilities arrival; incident log says response time was two minutes.	Flag discrepancy; do not replace the source value.
09:18	ICU team completed cold-chain relocation and logged chain of custody.	Proposed strength against OBJ-03.
Debrief	Radio follow-up action has no owner; separate note says generator lagged, without time or measure.	Owner remains blank; ambiguous note is not promoted.

The provided transcript includes a synthetic 0.58-confidence 'two' → 'too' recognition substitution. It stays uncertain and the incorrect-response-time draft cites the independent event log, not that transcript phrase.

3. AI and evidence architecture

The workflow is a fixed local pipeline: parse source → extract observations → detect candidate findings → map to a stated objective → attach exact observation IDs → surface timing/recognition uncertainty → draft an action → wait for a human decision. The same evidence IDs resolve to file name and row, line, page, or transcript segment.

Evidence rule
Every displayed candidate includes supporting evidence. If the source is incomplete, the gap is explicit. A model may not fill it with a plausible event or resolve a time conflict.

MVP technical choices

Layer	Selection	Rationale / boundary
Server and data checks	FastAPI 0.162.0 · Pydantic 2.14.1	Receives website requests and checks that uploaded data has the expected fields.
Saved data	SQLAlchemy 2.0.48 · SQLite default · PostgreSQL 18.6 Compose	One data-access path supports a local file database or a deployed server database.
Website screen	React 19.3.1 · TypeScript 5.9.3 · Vite 8.3.5	Pages people use to plan exercises, review evidence, and manage tasks.
PDF import	pypdf 6.21.2 (BSD-3-Clause)	Selectable text only; chosen over an AGPL/commercial-dual-license parser.
Speech-to-text (ASR)	faster-whisper 1.2.1 optional (MIT)	Runs locally when a model is installed; it creates a draft transcript and does not identify speakers.
Optional writing model (LLM)	Ollama + Qwen3-4B optional (Apache 2.0 model weights)	May reword an action draft only; not used in the recorded evaluation.
Reports	python-docx 1.2.0 · ReportLab 5.1.0 · PptxGenJS 4.0.1	Editable DOCX/PPTX; PDF report/export and speaker notes.
Quality	pytest 9.1.1 · Ruff 0.17.2	Parser, status transition, citation, and API tests.

Why fixed workflow instead of agents and retrieval

This MVP uses a short list of pre-written rules. The same input gives the same draft, so a reviewer can inspect why it appeared. The sample has only a few files, so a separate search system for similar text would add complexity without a clear benefit. A tool such as LangGraph could be considered later if teams need long-running review queues or several people must sign off in sequence.

4. Differentiation and competitive context

After-action report generation and follow-up tasks are already available in existing products. Veoci publicly describes timestamps, communication transcripts, reports, assigned tasks, and reminders. D4H serves emergency-response readiness and related team workflows. FEMA's Homeland Security Exercise and Evaluation Program (HSEEP) provides exercise-planning and evaluation guidance. A report template alone is not a unique product claim.

Option	Publicly visible scope	DrillLoop implication
FEMA HSEEP / Preparedness Toolkit	Federal exercise-planning and evaluation guidance, including after-action and improvement planning.	Use familiar concepts; do not imply official status or replace an organization's required reports.
Veoci	Emergency-management platform; after-action reporting, timestamping, records, and corrective-action follow-up.	Direct competitor and integration/market-fit question; evidence trace must be tested in product demos.
D4H	Emergency response team readiness and adjacent incident/personnel/equipment workflows.	Adjacent incumbent; confirm evaluator evidence-review and closure workflows with customers.
DrillLoop positioning hypothesis	Exact source observation → proposed finding → reviewer decision → closure evidence.	Potential niche: evidence-to-review trace, uncertainty, and action closure. Uniqueness is unvalidated.

Differentiator to validate

For each finding, the evaluator can open the exact source text at its filename and row/line/page/segment, see the claim separated from interpretation, confirm or reject it, and later inspect reviewer-approved closure evidence. The differentiator hypothesis is trustworthy traceability across the review lifecycle—not automated prose by itself.

Research references include a public hospital power-loss exercise scenario from California Department of Public Health and a public health-coalition exercise AAR/IP example. These support scenario familiarity and report format; they do not validate product demand.

5. Feasibility, security, and governance

Feasibility

The app runs without a cloud AI service. It uses SQLite for the local demo and can use PostgreSQL in Docker Compose. The website lets a user create an exercise, upload evidence, inspect proposed findings, review actions, and download a PDF report. The server checks uploads, runs the fixed analysis rules, and records reviewer decisions. Optional speech-to-text and local text-generation paths were not used in the benchmark.

Security boundary

- The demo uses one shared reviewer password. It is not a named-user login system and does not give different people different permissions.
- Original files and extracted text are saved in the configured database. The demo does not provide encryption at rest, file scanning, data-retention controls, scanned-page reading, or a security review.
- Do not upload real patient or operational data to the hackathon demo. Production needs approved hosting, encryption and key management, strict user/role controls, audit access, retention rules, dependency scanning, and threat modeling.
- No cloud model call is required. Optional Ollama must be installed locally; weights should be staged, versioned, license-checked, and egress-restricted before private deployment.

Human review and professional judgment

An authorized reviewer must approve or reject a proposed finding. An action needs an approved finding, owner, and due date before action approval. Closing it also requires reviewer confirmation and completion evidence. The report borrows concepts from FEMA's Homeland Security Exercise and Evaluation Program (HSEEP), but it is not FEMA-certified.

6. Evaluation, commercialization, and roadmap

Recorded technical benchmark

Measure	Observed	Interpretation
Observation extraction	8/8 · precision 1.00 · recall 1.00	Labeled CSV rows in one synthetic fixture.
Finding detection	5/5 · precision 1.00 · recall 1.00	Closed ground-truth finding set; not generalization.
Exact cited event IDs	9/9	All labeled supporting IDs were present.
Timeline row times	8/8 exact matches	Event-log rows only; conflicts remain surfaced separately.
Action register fields	31/32	One intentionally missing owner.
Rule route latency	p50 4.61 ms · p95 9.13 ms	Local TestClient call; no LLM or ASR inference.

Manual evaluator performance, report-preparation time, correction workload, ASR word error rate, local LLM quality, and monetary savings have not been measured. The results are not a field-accuracy claim.

Commercial hypotheses

Potential packaging to test: $1,500 per exercise for small users; $18,000 annual organization subscription including up to 12 exercises; $65,000/year private deployment; $6,000 custom template; and $1,500/day implementation or evaluator-support services. These are illustrative willingness-to-pay probes, not quoted market rates. Annual support, onboarding, security obligations, and procurement costs need validation.

A labor-only workload hypothesis assumes 20 documentation hours per exercise, 40% of those hours potentially reduced, 12 exercises per year, and $65 fully-loaded evaluator labor per hour. That yields 96 capacity hours or $6,240 of annual capacity value—below the example $18,000 subscription. Time saved does not equal cash savings; customers must validate both hours and other operational value.

Roadmap

Horizon	Proposed work	Evidence required to proceed
0–3 months	Pilot with 2–3 hospital/municipal exercise teams; measure evidence review and correction effort.	Evaluator agreement, citation accuracy, privacy approval, and time logging.
3–6 months	Role-based authentication, configurable templates, action-owner workflow, export tuning.	Customer fit, procurement requirements, independent report review.
6–12 months	Evaluate integrations, long-term trend views, and optional retrieval/model support.	Measured citation benefit, consented data, model license and bias review.

Key risks

Incomplete inputs; timestamp ambiguity; ASR errors; rules that overfit; confidential evidence exposure; weak adoption; existing platforms that already satisfy the need; and the risk that buyers pay only for convenience instead of measurable improvement. Pilot success criteria should emphasize source-supported findings, reviewer workload, action ownership, and independently verifiable closure.

7. Sources and review notes

The public references below informed workflow and positioning. Accessed or version-checked 26 October 2026. Competitor features are summarized from public product materials, not a scored product teardown.

[1] FEMA, Homeland Security Exercise and Evaluation Program Doctrine, January 2020: https://www.fema.gov/sites/default/files/2020-04/Homeland-Security-Exercise-and-Evaluation-Program-Doctrine-2020-Revision-2-2-25.pdf

[2] FEMA Preparedness Toolkit, Exercise Evaluation: https://preptoolkit.fema.gov/web/training/evaluation

[3] California Department of Public Health, Power Loss scenario and hospital objectives: https://www.cdph.ca.gov/Programs/EPO/Pages/swmhe_scenarios_power_loss.aspx

[4] Veoci, After-action reporting product page: https://veoci.com/emergency-management/after-action-reporting/

[5] D4H, Emergency response team readiness software: https://www.d4h.com/team-readiness-software

[6] Mid-South Emergency Planning Coalition, 2023 Full Scale Exercise AAR/IP example: https://midsouthepc.org/wp-content/uploads/2024/03/Mid-South-HCC-2023-Non-Hospital-FSE-AAR.pdf

[7] Python release history: https://www.python.org/doc/versions/ ; FastAPI 0.162.0: https://pypi.org/project/fastapi/0.162.0/

[8] React versions: https://react.dev/versions ; PostgreSQL versioning policy: https://www.postgresql.org/support/versioning/

[9] faster-whisper: https://github.com/SYSTRAN/faster-whisper ; MIT English-small model card: https://huggingface.co/Systran/faster-whisper-small.en ; Whisper license: https://github.com/openai/whisper/blob/main/README.md

[10] Qwen3 model licenses and local use: https://qwenlm.github.io/blog/qwen3/ ; Ollama local API: https://github.com/ollama/ollama/blob/main/docs/api.md

[11] pypdf 6.21.2 license metadata (BSD-3-Clause): https://pypi.org/project/pypdf/6.21.2/

[12] PptxGenJS speaker notes: https://gitbrent.github.io/PptxGenJS/docs/speaker-notes.html

Review disclaimer

DrillLoop provides a working, synthetic demonstration. It does not make safety-critical decisions or replace exercise evaluator judgment, organizational procedures, or jurisdictional reporting requirements.